

HAT-P-18b

R-2502

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-18_180620 · created 2026-08-24T23:21:33+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458289.73327 +/- 0.00019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1449 +/- 0.0079
Transit depth (Rp/Rs)^2	2.1 +/- 0.23 [%]
Orbital Inclination (inc)	89.02 +/- 0.46
Airmass coefficient 1 (a1)	1.088 +/- 0.012
Airmass coefficient 2 (a2)	-0.08 +/- 0.021
Scatter in the residuals of the lightcurve fit is	1.14 %
Best Comparison Star	None
Optimal Aperture	3.87
Optimal Annulus	13.02
Transit Duration (day)	0.1156 +/- 0.0037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1449 ± 0.0079 vs 0.1356 (deviation 1.13σ)
Duration fit vs published	0.1156 d vs 0.1129 d (deviation 0.69σ)
Mid-time O–C	-1.0 min
Depth SNR	17.5
Residual scatter	1.14 %

Light curve

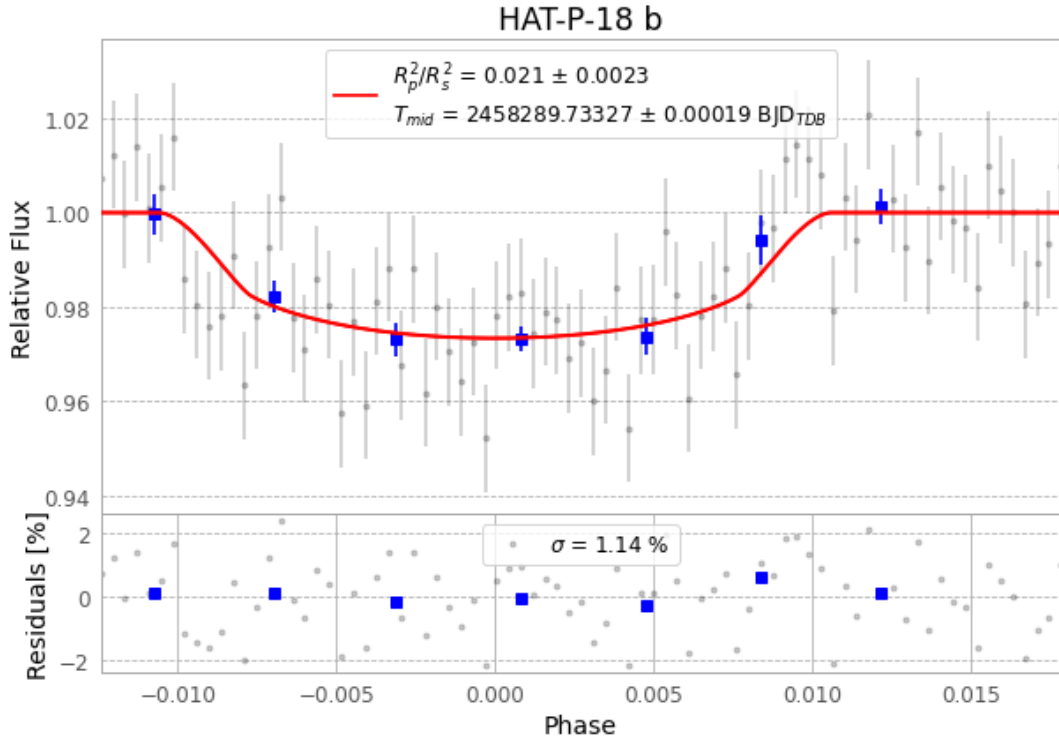


Figure 1 — FinalLightCurve_HAT-P-18 b_2018-06-19.png (SHA-256 ebaab247b2a75be6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [381, 201], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1a3e22a063c96b46...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

81 FITS frames (53 MB), HATP-18180620035112.FITS → HATP-18180620075112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-180620.log (75555bb71e9f...), FinalLightCurve_HAT-P-18 b_2018-06-19.png (ebaab247b2a7...), FinalLightCurve_HAT-P-18 b_2018-06-19.csv (e1ffcdb30b7a...)

Compute resources

Wall time 118.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 23:21Z.